Unit 2 - Linear Algebra
2.3 Matrices

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MSDS 520: Math and Stats Foundations for Data Science
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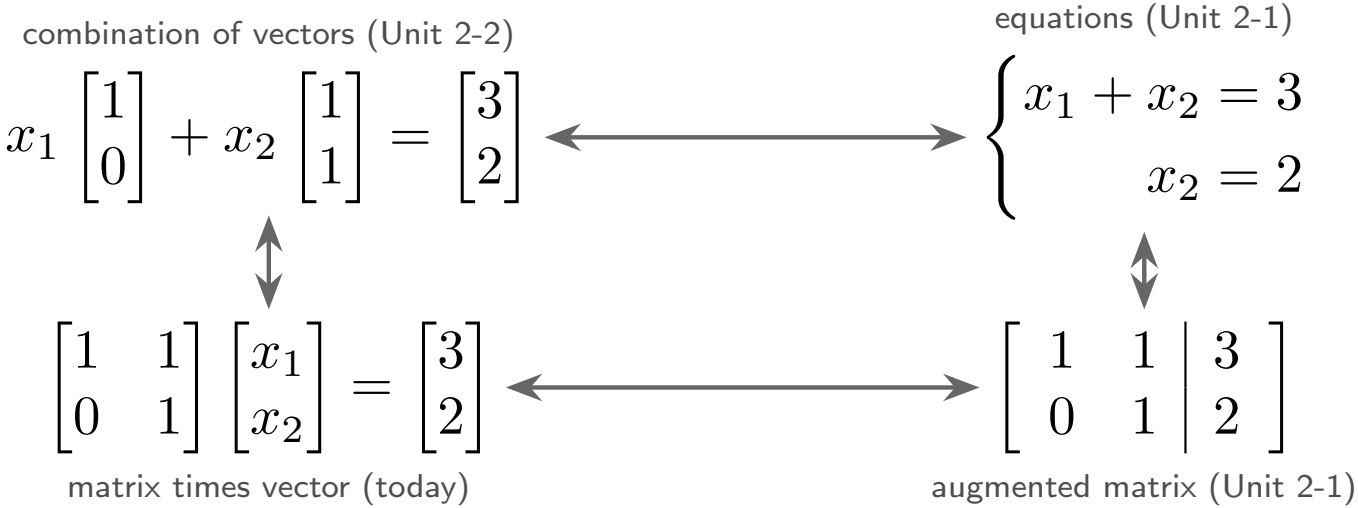
Outline

1. Where Vectors Left Off
2. Why Matrices?
3. Matrices and Their Entries
4. Matrix Arithmetic
5. Block Notation
6. Matrix Times Vector
7. Matrix Times Matrix
8. Networks
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Where Vectors Left Off

A Dependence Problem, Four Ways

Unit 2-2 ended here: is $\bar{u} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ a **combination** of $\bar{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\bar{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$? If so, $\bar{u}$ is redundant.



Naming the pieces: $A\bar{x} = \bar{b}$

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

The **coefficient matrix**. Its columns are $\bar{v}$ and $\bar{w}$.

$$\bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

The **unknowns**: how much of each column.

$$\bar{b} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

The **target**: the right-hand side, here $\bar{u}$.

$A\bar{x} = \bar{b}$ reads: **weight the columns of A by $\bar{x}$, and hit $\bar{b}$.**

Why Matrices?

A Problem to Motivate Us

A matinee, from Unit 2-1

"A theater sold 300 tickets: children's at \$$6$, adults' at \$$10$, seniors' at \$$8$, and took in \$$2540$. As many adult tickets were sold as children's and seniors' combined. How many of each?"

Unit 2-1: three equations.

$$\begin{cases} c + a + s = 300 \\ 6c + 10a + 8s = 2540 \\ -c + a - s = 0 \end{cases}$$

Today: **one** equation, $A\bar{x} = \bar{b}$.

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 6 & 10 & 8 \\ -1 & 1 & -1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} c \\ a \\ s \end{bmatrix}}_{\bar{x}} = \underbrace{\begin{bmatrix} 300 \\ 2540 \\ 0 \end{bmatrix}}_{\bar{b}}$$

Where we are headed

- A holds the **prices and rules**, $\bar{x}$ the unknown **counts**, $\bar{b}$ the **totals**.
- By the end of **Matrix Times Vector**, one line of code solves it.

Where This Shows Up in Data Science

Almost everything you compute with is a matrix

- **A data table is a matrix.** One row per record, one column per feature. Unit 2-2 read the rows as vectors; now the whole table is one object.
- **A network is a matrix.** Facebook friendships, web links, and gene interactions are all stored as a grid of who connects to whom.
- **Fitting a model is a matrix product.** Least-squares regression is built from $X^\top X$, a product you will be able to compute by the end of today.
- **Moving an image is matrix multiplication.** Classifiers train on rotated and flipped copies of every picture (**augmentation**), all made this way.
- **PCA**, this unit's capstone, starts from a **symmetric** covariance matrix and **rotates** the axes to fit the data.
- **A neural network layer** is a matrix transformation followed by a simple nonlinear step.

Unit 2-1 already used a grid of numbers, the augmented matrix. Today the grid becomes an object with its own arithmetic.

What We Will Build

This deck: the machinery

Matrices, their arithmetic, and their **products**.

- Mostly **technical**: definitions, notation, and the rules of matrix arithmetic.
- Little application or geometry, **on purpose**: systems, vectors, dot products, and sums all have to be joined into one tool first.
- Expect to **compute** more than to picture.

Unit 2-4: putting it to work

A matrix as a **function** that moves the plane.

- Much more **geometric**: a matrix moves the whole plane, and you can watch it happen.
- Much more **applicable**: rotations, projections, and the road to PCA.
- Much more **intuitive**: the rules from this deck finally get pictures.

If this deck feels dry, that is expected: Unit 2-4 puts the tool to work.
Notebooks: `U2-3_Matrices-1`, `-2`.

Matrices and Their Entries

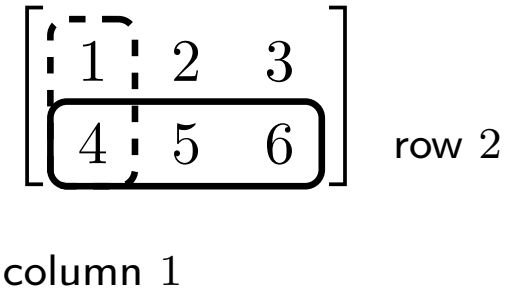
What Is a Matrix?

Definition

An $m \times n$ **matrix** is a rectangular grid of numbers with m **rows** and n **columns**.

$$A = \underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}}_{n = 3 \text{ columns}} \Bigg\} \begin{matrix} m = 2 \\ \text{rows} \end{matrix}$$

A 2×3 matrix.



A **row** runs across; a **column** runs down.

Size is always rows $\times$ columns

- Rows first, always: this A is 2×3, never 3×2.
- A 3×1 matrix is **tall**; a 1×3 matrix is **wide**.
- When $m = n$ the matrix is **square**.

Naming Entries

Every entry has an address:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

a_{ij} sits in **row** i, **column** j. The first subscript is the row, the second the column.

$$A = \begin{bmatrix} 1 & \textcircled{2} & 3 \\ 4 & 5 & \textcircled{6} \end{bmatrix}$$

$a_{12} = 2$

$a_{23} = 6$

Row 1, column 2; row 2, column 3.

The naming convention: the letter tells you what it is

- **Capital letter: a matrix.** A, B, M.
- **Same letter, lowercase, two subscripts: its entries.** a_{ij} lives in A, b_{ij} in B.
- **Lowercase with a bar: a vector.** $\bar{v}$, $\bar{x}$, as in Unit 2-2.
- **Plain lowercase: a number** (a **scalar**). c, k.

A Vector Is a Matrix Too

The column vectors of Unit 2-2 are matrices with **one column**. A vector can also be written as a single row: a matrix with **one row**.

$$\bar{v} = \underbrace{\begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix}}_{1 \text{ column}} \Big\} 3 \text{ rows}$$

Column vector: 3×1, **tall**.

$$\bar{w} = \underbrace{\begin{bmatrix} 2 & 7 & 5 \end{bmatrix}}_{3 \text{ columns}} \Big\} 1 \text{ row}$$

Row vector: 1×3, **wide**.

Two shapes of vector

- The **column** is the default: unless we say otherwise a vector is a column, so $\bar{v} \in \mathbb{R}^n$ is $n \times 1$.
- A **row** vector such as $\bar{w}$ is $1 \times n$. Rows meet columns when we multiply, later today.
- Every rule for matrices today applies to vectors too. Nothing new to learn.

When Are Two Matrices Equal?

Equal means: same size, and every matching entry agrees

For vectors and matrices alike: check the **size** first, then $a_{ij} = b_{ij}$ for all i, j.

	Vectors	**Matrices**
Equal	$\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
Not equal: same numbers, wrong places	$\begin{bmatrix} 3 \\ 4 \end{bmatrix} \neq \begin{bmatrix} 4 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \neq \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$
Not equal: sizes differ, so stop there	$\begin{bmatrix} 3 \\ 4 \end{bmatrix} \neq \begin{bmatrix} 3 & 4 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \neq \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

A row is never a column, whatever the numbers. And one matrix equation is one equation per entry: $\begin{bmatrix} x & 2 \\ 3 & y \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix}$ means $x = 5$, $y = -1$.

Matrices in Python

A matrix is a 2-D NumPy array

```
import numpy as np
A = np.array([[1, 2, 3],
              [4, 5, 6]])
A.shape   # (2, 3): rows, columns
A[0, 1]   # 2, which is a_12
A[1, 2]   # 6, which is a_23
A[0, :]   # row 1:    [1 2 3]
A[:, 1]   # column 2: [2 5]
```

Python counts from 0

Mathematics counts from 1, Python from 0:

a_{ij} is `A[i-1, j-1]`

This off-by-one is the most common bug in a first linear algebra notebook.

Vectors: a 1-D array has no shape until you give it one

```
v = np.array([3, 4, 1])   # shape (3,):   no row or column
v.reshape(3, 1)           # shape (3, 1): a column, 3 x 1
v.reshape(1, 3)           # shape (1, 3): a row,    1 x 3
```

Notebook `U2-3_Matrices-1_ArithmeticAndSpecial`, §1, entries and dimensions

A Data Table Is a Matrix

Three patients, three measurements (the first two are Unit 2-2's $\bar{p}$ and $\bar{q}$):

	Age	**BP**	**A1c**
Patient 1	54	130	6.1
Patient 2	58	138	6.4
Patient 3	41	118	5.6

$$\xrightarrow{\text{numbers only}} \quad X = \begin{bmatrix} 54 & 130 & 6.1 \\ 58 & 138 & 6.4 \\ 41 & 118 & 5.6 \end{bmatrix}$$

The same grid, two views

- Each **row** is one patient: row 1 is patient 1's measurements.
- Each **column** is one feature across everyone: age is a vector in $\mathbb{R}^3$.
- x_{ij} is feature j of patient i: $x_{23} = 6.4$ is patient 2's A1c.

Both views matter

- Similar **patients**: compare rows. Redundant **features**: compare columns.
- One sample per row is how data is **stored**; a lone vector is still a column.

Exercise: Read the Matrix

Try it

Let

$$M = \begin{bmatrix} 3 & 0 & -1 \\ 2 & 7 & 4 \end{bmatrix}, \qquad S = \begin{bmatrix} 2 & 5 \\ 5 & -1 \end{bmatrix}.$$

1. State the dimensions of M, and read off m_{13} and m_{21}.
2. Can M ever equal S? Why or why not?
3. A patient record with 12 lab values is stored for 250 patients. What are the dimensions of the data matrix, and what does the entry in row 17, column 3 mean?

Exercise: Solution

Answers

1. M is 2×3: two rows, three columns. m_{13} is row 1, column 3, which is -1; m_{21} is row 2, column 1, which is 2. (Not the other way round: rows first.)
2. **Never.** M is 2×3 and S is 2×2. Equality requires the same shape before a single entry is compared.
3. 250×12: one row per patient, one column per lab value. Row 17, column 3 is the **third lab value of the seventeenth patient**. The row is that patient's feature vector; the column is that lab value across all 250 patients.

Matrix Arithmetic

Addition and Scalar Multiplication

Entry by entry, exactly as for vectors

Matrices of the **same shape** add entry by entry, and a scalar multiplies every entry:

$$(A+B)_{ij} = a_{ij} + b_{ij}, \qquad (cA)_{ij} = c\,a_{ij}.$$

Subtraction is $A - B = A + (-1)B$; vectors are the one-column case.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}, \qquad 2\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

Rules from numbers

- $A + B = B + A$
- $(A + B) + C = A + (B + C)$
- $c(A + B) = cA + cB$, and $A + 0 = A$

Watch out: the size rule from equality

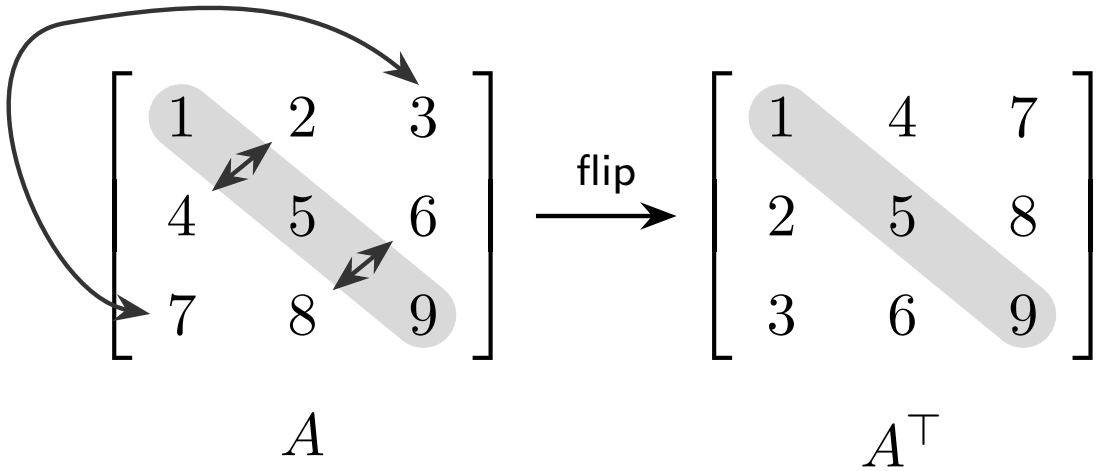
Adding and subtracting need the **same size**, just like equality. A 2×2 plus a 2×3 is **undefined**, and so is a column plus a row, $\left[\begin{smallmatrix} 3 \\ 4 \end{smallmatrix}\right] + [\,3\ 4\,]$. `numpy` errors on the first but silently **broadcasts** the second to a 2×2 grid.

Notebook `U2-3_Matrices-1_ArithmeticAndSpecial`, §2, sums and scalar multiples

The Transpose

Definition: transpose

The **transpose** $A^\top$ swaps rows and columns: $(A^\top)_{ij} = a_{ji}$. The grid flips over its **main diagonal** $(a_{11}, a_{22}, \ldots)$, so an $m \times n$ matrix becomes $n \times m$.



A $\qquad$ $A^\top$

The diagonal (shaded) stays put; every other entry swaps with its mirror image.

Not square? The shape flips too:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^\top = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

Rules

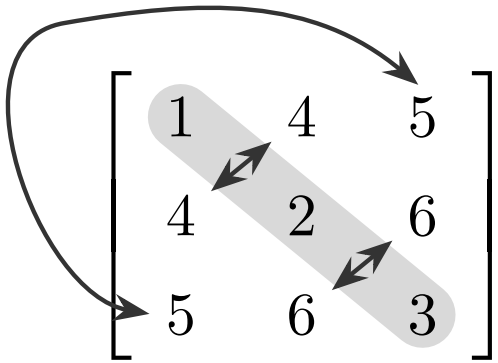
$$(A^\top)^\top = A, \quad (A+B)^\top = A^\top + B^\top, \quad (cA)^\top = c\,A^\top.$$

Notebook `U2-3_Matrices-1_ArithmeticAndSpecial`, §3, the transpose

Symmetric Matrices

Definition: symmetric

A square matrix A is **symmetric** when $A^\top = A$, that is, $a_{ij} = a_{ji}$ for every i and j: a **mirror image** across the diagonal, so the flip changes nothing.



$$S = S^\top$$

Every pair that trades places holds the same number.

Worth knowing

- Only a **square** matrix can be symmetric: the flip has to keep the size.
- **Not** symmetric: $\left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$, since $a_{12} = 2$ but $a_{21} = 3$.
- The **covariance matrix** behind PCA is symmetric, and so is a **friendship** matrix (later today).

Exercise: Arithmetic and Transposes

Try it

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 \\ 2 & 5 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 0 \end{bmatrix}$.

1. Compute $A + B$ and $3A - B$.
2. Write down $A^\top$ and $C^\top$, with their dimensions.
3. Compute $A + A^\top$. Is it symmetric? Would that be true for **any** square A?
4. What is wrong with $A + C$?

Exercise: Solution

Answers

1. $A + B = \begin{bmatrix} 1 & 1 \\ 5 & 9 \end{bmatrix}$. $3A = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}$, so $3A - B = \begin{bmatrix} 3 & 7 \\ 7 & 7 \end{bmatrix}$. Scale first, then subtract entry by entry.
2. $A^\top = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$, still 2×2. $C^\top = \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ 2 & 0 \end{bmatrix}$, now 3×2: the rows of C have become its columns.
3. $A + A^\top = \begin{bmatrix} 2 & 5 \\ 5 & 8 \end{bmatrix}$, which **is** symmetric. And it always is: $(A + A^\top)^\top = A^\top + (A^\top)^\top = A^\top + A$, the same matrix. Adding a square matrix to its own transpose is the standard way to **manufacture** a symmetric matrix.
4. **The shapes differ.** A is 2×2 and C is 2×3; there is no third column of A to add to the third column of C. The sum is undefined.

Block Notation

Block Notation: a Matrix by Its Columns or Its Rows

The same matrix can be grouped into vectors in two ways:

By columns

$$\left[\begin{array}{cc} \boxed{\begin{matrix}1\\3\end{matrix}} & \boxed{\begin{matrix}2\\4\end{matrix}} \end{array}\right]$$

$\bar{c}_1 \qquad \bar{c}_2$

$$A = \begin{bmatrix} \bar{c}_1 & \bar{c}_2 \end{bmatrix}, \quad \bar{c}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \; \bar{c}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

By rows

$$\left[\begin{array}{cc} \boxed{\begin{matrix}1 & 2\end{matrix}} \\ \boxed{\begin{matrix}3 & 4\end{matrix}} \end{array}\right] \begin{matrix} \bar{r}_1^\top \\ \bar{r}_2^\top \end{matrix}$$

$$A = \begin{bmatrix} \bar{r}_1^\top \\ \bar{r}_2^\top \end{bmatrix}, \quad \bar{r}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \; \bar{r}_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

What the notation means

- The brackets now hold **vectors**, not numbers: this is **block notation**.
- A vector is always a **column**, so row i is written as a column **transposed**: $\bar{r}_i^\top$, $1 \times n$.
- Any size works: an $m \times n$ matrix is n columns side by side, $\begin{bmatrix} \bar{c}_1 & \bar{c}_2 & \cdots & \bar{c}_n \end{bmatrix}$, or m rows stacked.

Block Notation: Blocks That Are Matrices

Blocks can be matrices too. Cut a 4×4 matrix down the middle, both ways:

$$M = \begin{bmatrix} \overbrace{\begin{matrix} 1 & 2 \\ 5 & 6 \end{matrix}}^{M_{11}} & \overbrace{\begin{matrix} 3 & 4 \\ 7 & 8 \end{matrix}}^{M_{12}} \\ \underbrace{\begin{matrix} 9 & 10 \\ 13 & 14 \end{matrix}}_{M_{21}} & \underbrace{\begin{matrix} 11 & 12 \\ 15 & 16 \end{matrix}}_{M_{22}} \end{bmatrix}$$

Four 2×2 blocks

$$M = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix}$$

$$M_{11} = \begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix}, \quad M_{12} = \begin{bmatrix} 3 & 4 \\ 7 & 8 \end{bmatrix}$$

$$M_{21} = \begin{bmatrix} 9 & 10 \\ 13 & 14 \end{bmatrix}, \quad M_{22} = \begin{bmatrix} 11 & 12 \\ 15 & 16 \end{bmatrix}$$

The rules for blocks

- M_{12} is the block in block-row 1, block-column 2: **rows first**, as for entries.
- Each cut runs **all the way across**, so blocks side by side share a height and blocks stacked share a width.
- Columns and rows are the special case: blocks of size $m \times 1$ or $1 \times n$.

Matrix Times Vector

Where the Product Comes From

The question (Unit 2-2)

What weights c_1, c_2 reach $\bar{p} = \left[\begin{smallmatrix} 3 \\ 3 \end{smallmatrix}\right]$? As a combination, or as a system:

$$c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2c_1 - c_2 \\ c_1 + c_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \quad \longleftrightarrow \quad \begin{cases} 2c_1 - c_2 = 3 \\ c_1 + c_2 = 3 \end{cases}$$

The fix: give the pieces names

Both forms scatter the unknowns. Instead, the basis vectors become the **columns** of B, the weights stack into $\bar{c}$, and $B\bar{c}$ is **defined** as that combination:

$$\underbrace{\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}}_{B} \underbrace{\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}}_{\bar{c}} = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Not Multiplication as You Know It

In arithmetic, multiplying takes two numbers and gives a number (the **same kind** in and out), and **order does not matter**: $3 \times 4 = 4 \times 3$.

Linear algebra's "products" break those habits

	In	**Out**	**Swap the order?**
3×4	two numbers	a number	same
$A\bar{x}$	$m \times n$, $n \times 1$	$m \times 1$	**undefined**
$\bar{v} \cdot \bar{w}$	two vectors	a **number**	same
AB (soon)	two $n \times n$	$n \times n$	usually **different**

Watch out

- A vector **is** a matrix ($n \times 1$), so A and $\bar{x}$ are the same kind of thing, but their **shapes** must fit.
- "Product" is a name **by analogy**: built from ordinary $\times$ and $+$, but arithmetic's rules do **not** come along.

Matrix Times Vector: the Definition

Definition

A $(m \times n)$ times $\bar{x}$ $(n \times 1)$ is the $m \times 1$ vector with entries $(A\bar{x})_i = \sum_{j=1}^{n} a_{ij}\, x_j$.

For 2×2:

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{bmatrix}$$

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix},\ \bar{x} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

Row view: how you **compute**

Group the terms **across each row**:

$$A\bar{x} = \begin{bmatrix} \bar{r}_1^{\top} \\ \bar{r}_2^{\top} \end{bmatrix} \bar{x} = \begin{bmatrix} \bar{r}_1 \cdot \bar{x} \\ \bar{r}_2 \cdot \bar{x} \end{bmatrix}$$

$$\begin{bmatrix} 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 5 + 4 \cdot 6 \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}$$

Column view: how you **think**

Group the terms **by x_1 and x_2**:

$$A\bar{x} = \begin{bmatrix} \bar{c}_1 & \bar{c}_2 \end{bmatrix} \bar{x} = x_1\bar{c}_1 + x_2\bar{c}_2$$

$$5 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 6 \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}$$

Matrix Times Vector: Worked, and Impossible

Worked: $(2 \times 2)(2 \times 1)$

$$\begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \cdot 3 + 1 \cdot 2 \\ 4 \cdot 3 + 3 \cdot 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 18 \end{bmatrix}$$

Worked: $(2 \times 3)(3 \times 1)$, not square

$$\begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 + 0 + 6 \\ 3 + 2 + 3 \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

Sizes must match, or there is no product

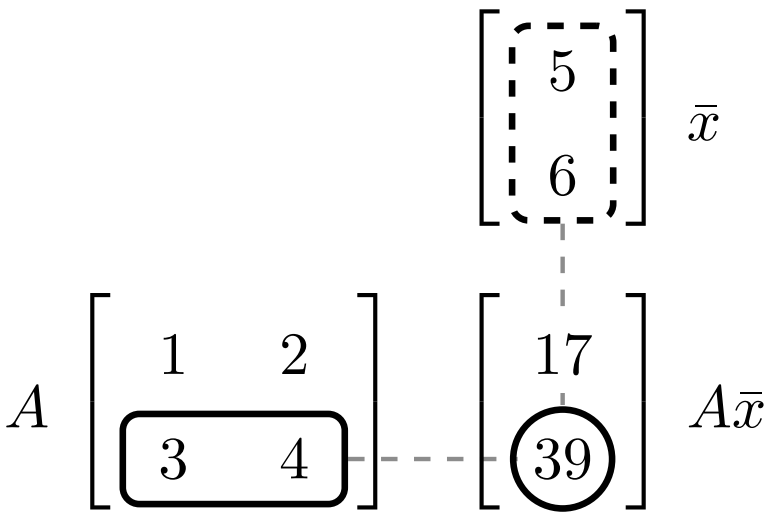
$(m \times \underbrace{n}_{\text{equal}})\ (\underbrace{n}_{\text{equal}} \times 1) = (m \times 1)$: a row of A must be **exactly as long** as $\bar{x}$.

Impossible:

- $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is $(2 \times \mathbf{2})(\mathbf{3} \times 1)$: each row has 2 entries, $\bar{x}$ has 3. The 3 has nothing to pair with.
- $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is $(2 \times \mathbf{3})(\mathbf{2} \times 1)$: each row has 3 entries, $\bar{x}$ only 2.

Undefined means undefined: no zero-padding, no partial answer. `numpy` raises an error.

Line It Up: Where Each Entry of $A\bar{x}$ Goes



How to use it

- Put $\bar{x}$ **above** and A to the **left**; the answer $A\bar{x}$ fills the corner.
- Row i of A runs **across**, $\bar{x}$ runs **down**, and they meet at entry i of $A\bar{x}$.
- That entry is their dot product: row 2 gives $3 \cdot 5 + 4 \cdot 6 = 39$.

The layout shows the shape rule too: a row of A must be exactly as long as $\bar{x}$, or the pairs do not line up.

Side by Side Now Means Something

Notation, updated from Unit 2-2

Written	**Shapes**	**Meaning**
$A\bar{x}$	$(m \times n)(n \times 1)$	matrix times vector: a vector in $\mathbb{R}^m$
$\bar{x}A$	$(n \times 1)(m \times n)$	**undefined**: the inner sizes do not match
$\bar{y}^\top A$	$(1 \times m)(m \times n)$	a **row**, $1 \times n$: $\bar{y}$ dotted with each column
$\bar{v}\,\bar{w}$	$(n \times 1)(n \times 1)$	still **undefined** for two columns
$\bar{v}^\top \bar{w}$	$(1 \times n)(n \times 1)$	a **number**, 1×1: the dot product $\bar{v} \cdot \bar{w}$
$\bar{v}\,\bar{w}^\top$	$(n \times 1)(1 \times n)$	an $n \times n$ **matrix** (outer product, Unit 2-4)

The dot product is a matrix product

A row $\bar{v}^\top$ is a column tipped over, $1 \times n$. Row times column is 1×1, a number:

$$\bar{v}^\top \bar{w} = \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = v_1 w_1 + v_2 w_2 = \bar{v} \cdot \bar{w}, \qquad \text{e.g. } \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 3 + 8 = 11$$

So each entry of $A\bar{x}$ is a row of A times $\bar{x}$: $(A\bar{x})_i = \bar{r}_i^\top \bar{x}$.

A System Is Now One Equation

Back to the matinee. The augmented matrix of Unit 2-1 was this product in disguise:

$$\begin{cases} c + a + s = 300 \\ 6c + 10a + 8s = 2540 \\ -c + a - s = 0 \end{cases} \quad \longleftrightarrow \quad \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 6 & 10 & 8 \\ -1 & 1 & -1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} c \\ a \\ s \end{bmatrix}}_{\bar{x}} = \underbrace{\begin{bmatrix} 300 \\ 2540 \\ 0 \end{bmatrix}}_{\bar{b}}$$

Solve it in one line

- `np.linalg.solve(A, b)` gives $c = 80$, $a = 150$, $s = 70$.
- $[\,A \mid \bar{b}\,]$ from Unit 2-1 is just A with $\bar{b}$ pinned on.

Check it with the row view

$$A \begin{bmatrix} 80 \\ 150 \\ 70 \end{bmatrix} = \begin{bmatrix} 80 + 150 + 70 \\ 480 + 1500 + 560 \\ -80 + 150 - 70 \end{bmatrix} = \begin{bmatrix} 300 \\ 2540 \\ 0 \end{bmatrix} = \bar{b}$$

Application: A Bill in One Product

Two students take credits at two colleges, each with a fixed price per credit:

Credits taken

	College 1	College 2
Student 1	10	7
Student 2	11	4

Cost per credit

	\$ per credit
College 1	60
College 2	80

Credits are a matrix, prices a vector, and their product is every bill

$$\underbrace{\begin{bmatrix} 10 & 7 \\ 11 & 4 \end{bmatrix}}_{\text{credits}} \quad \underbrace{\begin{bmatrix} 60 \\ 80 \end{bmatrix}}_{\text{cost per credit}} = \underbrace{\begin{bmatrix} 1160 \\ 980 \end{bmatrix}}_{\text{bills}}$$

- **Row view:** student 1 pays $10(60) + 7(80) = 1160$.
- **Column view:** $60\begin{bmatrix} 10 \\ 11 \end{bmatrix} + 80\begin{bmatrix} 7 \\ 4 \end{bmatrix}$, price times credits, college by college.

The sizes match for a reason: the credits table's **columns** and the price vector's **entries** are the same two colleges.

Notebook `U2-3_Matrices-2_MultiplicationAndPowers`, §4, the bills computed

Special Matrices to Know on Sight

Zero matrix 0: every entry is 0, in any size

$$\underset{2\times 2}{\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}} \quad \underset{2\times 3}{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}} \quad \underset{3\times 1}{\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}$$

$$0\,\bar{x} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \bar{0}$$

Every row is all zeros, so the answer is $\bar{0}$. Also $0A = A0 = 0$: it **wipes out** anything, like 0.

Identity I: 1s on the diagonal, 0 elsewhere; always square

$$\underset{2\times 2}{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}} \quad \underset{3\times 3}{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$$

$$I\bar{x} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} = \bar{x}$$

Each row of I picks out one entry of $\bar{x}$. Also $IA = AI = A$: it **changes nothing**, like 1.

Notebook `U2-3_Matrices-1_ArithmeticAndSpecial`, §4, the special types displayed

Exercise: Matrix Times Vector

Work with these

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 3 & -1 \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\bar{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad \bar{v} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \quad \bar{w} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Try it

1. Compute $A\bar{u}$ by the row view, then again by the column view.
2. Which of $B\bar{u}$, $B\bar{v}$, $C\bar{u}$, $C\bar{v}$ exist? Compute the ones that do.
3. Compute $D\bar{w}$, $I\bar{w}$, and $0\,\bar{w}$, and say in words what each matrix did.
4. Compute $A^\top \bar{u}$. Is it the same as $A\bar{u}$?
5. Write $2x + 3y = 9,\ x + 4y = 7$ as $A\bar{x} = \bar{b}$, then check that $\bar{w}$ solves it.

Exercise: Solution

Answers

1. Rows: $A\bar{u} = \begin{bmatrix} 2(2)+3(-1) \\ 1(2)+4(-1) \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. Columns: $2\begin{bmatrix} 2 \\ 1 \end{bmatrix} - 1\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 4-3 \\ 2-4 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. They agree, as they must.
2. B is 2×3, so it needs a vector with 3 entries: $B\bar{v} = \begin{bmatrix} 1(1)+0(0)+2(2) \\ -1(1)+3(0)+1(2) \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$, and $B\bar{u}$, $(2 \times \mathbf{3})(\mathbf{2} \times 1)$, is **undefined**.

   C is 3×2, so it needs 2 entries: $C\bar{u} = \begin{bmatrix} 1(2)+2(-1) \\ 0(2)+1(-1) \\ 3(2)+(-1)(-1) \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 7 \end{bmatrix}$, a vector in $\mathbb{R}^3$ from one in $\mathbb{R}^2$. $C\bar{v}$, $(3 \times \mathbf{2})(\mathbf{3} \times 1)$, is **undefined**.

Exercise: Solution (continued)

Answers

3. $D\bar{w} = \begin{bmatrix} 0(3)+1(1) \\ 1(3)+0(1) \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$: D **swapped the two entries**, as it does to any vector, $D\begin{bmatrix} x \\ y \end{bmatrix} = x\begin{bmatrix} 0 \\ 1 \end{bmatrix} + y\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix}$. $I\bar{w} = \bar{w}$: the identity **left** $\bar{w}$ **alone**. $0\,\bar{w} = \bar{0}$: the zero matrix **sent** $\bar{w}$ **to the origin**.

4. $A^\top = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$, so $A^\top \bar{u} = \begin{bmatrix} 4-1 \\ 6-4 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$. **Different** from $A\bar{u}$: transposing changes which numbers sit in each row, so it changes the product.

5. $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 9 \\ 7 \end{bmatrix}$: the coefficients are A, the unknowns $\bar{x}$, the right-hand sides $\bar{b}$. Check: $A\bar{w} = \begin{bmatrix} 2(3)+3(1) \\ 1(3)+4(1) \end{bmatrix} = \begin{bmatrix} 9 \\ 7 \end{bmatrix} = \bar{b}$, so $x = 3$, $y = 1$ **solves** the system.

Matrix Times Matrix

Matrix Times Matrix: the Definition

Nothing new: matrix times vector, once per column

Write B by its columns, multiply A by each, and stand the answers side by side:

$$AB = A\begin{bmatrix}\bar{b}_1 & \bar{b}_2 & \cdots & \bar{b}_p\end{bmatrix} = \begin{bmatrix}A\bar{b}_1 & A\bar{b}_2 & \cdots & A\bar{b}_p\end{bmatrix}$$

Definition

For A $(m \times n)$ and B $(n \times p)$, entry (i, j) of AB is **row i of A dotted with column j of B**:

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik}\, b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

Sizes must match, or there is no product

$(m \times \underbrace{n}_{\text{equal}})\,(\underbrace{n}_{\text{equal}} \times p) = (m \times p)$: a row of A must be **as long as** a column of B.

Matrix Times Matrix: See Where Everything Goes

2×2 times 2×2, every entry written out

$$\underset{2\times 2}{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}} \underset{2\times 2}{\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}} = \underset{2\times 2}{\begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}}$$

i picks the row of A, j the column of B; k walks **across** and **down**.

Check: column 1 of the answer is just A times column 1 of B

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} \\ a_{21}b_{11} + a_{22}b_{21} \end{bmatrix}$$

A matrix-vector product, exactly as before. Column 2 is A times column 2 of B the same way.

Matrix Times Matrix: Worked, and Impossible

Worked: $(2 \times 2)(2 \times 2)$, every entry one dot product

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 7 & 1 \cdot 6 + 2 \cdot 8 \\ 3 \cdot 5 + 4 \cdot 7 & 3 \cdot 6 + 4 \cdot 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Worked: $(2 \times 3)(3 \times 2)$

$$\begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$$

Not square, but $3 = 3$: fine.

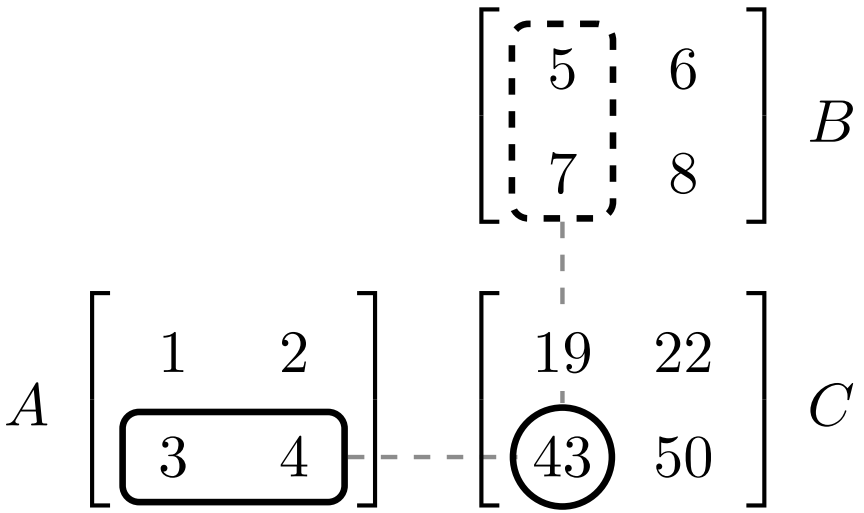
Impossible: the inner sizes do not match

- $(2 \times \mathbf{2})(\mathbf{3} \times 2)$: rows of A have 2 entries, columns of B have 3.
- $(2 \times \mathbf{3})(\mathbf{2} \times 3)$: a 2×3 matrix cannot even multiply **itself**.

No zero-padding, no partial answer; `numpy` raises an error.

Notebook `U2-3_Matrices-2_MultiplicationAndPowers`, §2, the top-left entry checked by hand

Line It Up: Where Each Entry of $AB = C$ Goes



How to use it

- Put B **above** and A to the **left**; $C = AB$ fills the corner.
- Row i of A runs **across**, column j of B runs **down**, and they cross at c_{ij}.
- c_{21} is row 2 dotted with column 1: $3 \cdot 5 + 4 \cdot 7 = 43$.

The layout explains the shapes

C gets A**'s rows** and B**'s columns**, and a row of A must be as long as a column of B: $(m \times n)(n \times p) = (m \times p)$. The matrix-vector picture is the case where B has one column.

Rules, and the One That Fails

What carries over from numbers

- **Associative:** $(AB)C = A(BC)$
- **Distributive:** $A(B + C) = AB + AC$
- **Identity:** $IA = AI = A$
- **Zero:** $0A = A0 = 0$
- **Transpose of a product:** $(AB)^\top = B^\top A^\top$, the order **reverses**

What does not: $AB \neq BA$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix} \neq \begin{bmatrix} 23 & 34 \\ 31 & 46 \end{bmatrix} = BA$$

The shapes can differ too

- $(2 \times 3)(3 \times 2) = 2 \times 2$, but $(3 \times 2)(2 \times 3) = 3 \times 3$.
- $(2 \times 3)(2 \times 3)$: **undefined**, in either order.

The habit to build: never swap the order of a matrix product without a reason.

Notebook `U2-3_Matrices-2_MultiplicationAndPowers`, §3, AB and BA compared

Why $AB \neq BA$

The top-left entry of each, with $A = \left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$ and $B = \left[\begin{smallmatrix} 5 & 6 \\ 7 & 8 \end{smallmatrix}\right]$

$$(AB)_{11} = \underbrace{\begin{bmatrix} 1 & 2 \end{bmatrix}}_{\text{row 1 of } A} \underbrace{\begin{bmatrix} 5 \\ 7 \end{bmatrix}}_{\text{column 1 of } B} = 1 \cdot 5 + 2 \cdot 7 = 19$$

$$(BA)_{11} = \underbrace{\begin{bmatrix} 5 & 6 \end{bmatrix}}_{\text{row 1 of } B} \underbrace{\begin{bmatrix} 1 \\ 3 \end{bmatrix}}_{\text{column 1 of } A} = 5 \cdot 1 + 6 \cdot 3 = 23$$

In general

$$(AB)_{ij} = \sum_k a_{ik}\, b_{kj} \qquad \text{vs.} \qquad (BA)_{ij} = \sum_k b_{ik}\, a_{kj}$$

- AB: rows of A with columns of B. BA: rows of B with columns of A.
- Different numbers get paired, so the sums differ. For numbers, ab and ba pair the **same** two.

Geometrically, in Unit 2-4: rotate then flip is not flip then rotate.

Matrix Powers

Definition: a square matrix times itself

$$A^2 = AA, \qquad A^3 = AAA, \qquad A^n = \underbrace{A\,A\cdots A}_{n \text{ times}}$$

Only square matrices have powers: $(m \times n)(m \times n)$ needs $n = m$.

Worked: $A = \left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$

$$A^2 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1+6 & 2+8 \\ 3+12 & 6+16 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}$$

Watch out

- A^2 is **not** each entry squared: $\left[\begin{smallmatrix} 1 & 4 \\ 9 & 16 \end{smallmatrix}\right]$ is wrong. In `numpy`, `A**2` squares entries; use `A @ A`.
- Powers of **one** matrix commute, $A^2A^3 = A^3A^2 = A^5$, but $(AB)^2 \neq A^2B^2$ in general.

Exercise: Matrix Products

Try it

Let $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{bmatrix}$.

1. Compute AB and BA. Are they equal?
2. State the dimensions of AC and compute it.
3. Explain why CA does not exist.
4. Compute $(AB)^\top$ and $B^\top A^\top$ and confirm they match.
5. Compute B^2. Is it the same as squaring each entry of B?

Exercise: Solution

Answers

1. $AB = \begin{bmatrix} 1(3)+0(0) & 1(1)+0(2) \\ 2(3)+1(0) & 2(1)+1(2) \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 6 & 4 \end{bmatrix}$,
   $BA = \begin{bmatrix} 3+2 & 0+1 \\ 0+4 & 0+2 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix}$. **Not equal.**
2. A is 2×2 and C is 2×3, so AC is 2×3: $AC = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & 3 \end{bmatrix}$. (Row 2: $2\begin{bmatrix} 1 & 2 & 0 \end{bmatrix} + 1\begin{bmatrix} 0 & 1 & 3 \end{bmatrix}$.)
3. CA would be $(2 \times 3)(2 \times 2)$. The inner dimensions are 3 and 2: a row of C has three entries and a column of A has two, so there is nothing to pair the third entry with. **Undefined.**

Exercise: Solution (continued)

Answers

4. $(AB)^\top = \begin{bmatrix} 3 & 6 \\ 1 & 4 \end{bmatrix}$. $\quad B^\top A^\top = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 1 & 4 \end{bmatrix}$. They match, and $A^\top B^\top$ would not: the order reverses under a transpose.
5. $B^2 = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 9+0 & 3+2 \\ 0+0 & 0+4 \end{bmatrix} = \begin{bmatrix} 9 & 5 \\ 0 & 4 \end{bmatrix}$. Squaring each entry gives $\left[\begin{smallmatrix} 9 & 1 \\ 0 & 4 \end{smallmatrix}\right]$: **different**, because the top-right entry of B^2 mixes a row with a column.

Networks

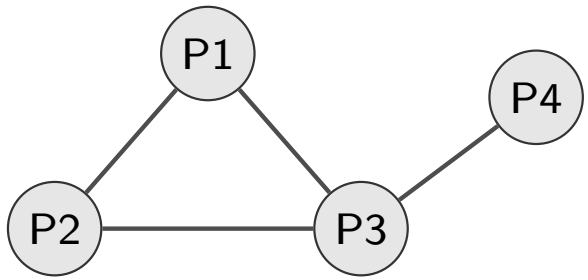
The Adjacency Matrix

Facebook friends: P1 is friends with P2 and P3, P2 with P3, and P3 with P4.

Definition

Draw each person as a node and each friendship as a line. The **adjacency matrix** records it: $a_{ij} = 1$ if i and j are friends, 0 if not.

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$



What the matrix shows

- **Symmetric**: a Facebook friendship is mutual. Diagonal 0: nobody is their own friend.
- Each **row sum** is a person's number of friends, their **degree**: $2, 2, 3, 1$, so P3 is the hub.

Notebook `U2-3_Matrices-2_MultiplicationAndPowers`, §5, the network beside its matrix

Squaring the Matrix Counts Two-Step Routes

For an adjacency matrix the product rule gives $(A^2)_{ij} = \sum_k a_{ik}\, a_{kj}$, and the term $a_{ik}a_{kj}$ is 1 exactly when k is friends with i **and** with j. Summing over k counts the people in between: $(A^2)_{ij}$ **is the number of two-step routes from** i **to** j, which for $i \neq j$ is their number of **mutual friends**.

$$A^2 = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 3 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

Check one entry: row 1 of A is $\begin{bmatrix} 0 & 1 & 1 & 0 \end{bmatrix}$ and column 4 is $\begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}^\top$, so $(A^2)_{14} = 0 + 0 + 1 + 0 = 1$: one mutual friend, P3.

Friend suggestions

Row 1 of A^2 is P1's mutual-friend counts: **one each** with P2 (via P3), with P3 (via P2), and with P4 (via P3). P4 is not P1's friend, yet they share a friend: exactly how "People You May Know" finds P4 for P1.

The diagonal of A^2 is each person's number of friends (out to a friend and straight back), and $(A^2)_{34} = 0$: P3 and P4 are friends but have no mutual friend.

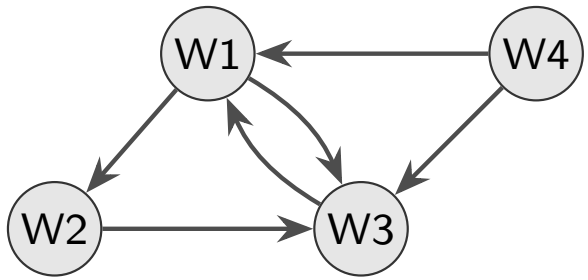
A Directed Network: Website Links

Four web pages: W1 links to W2 and W3, W2 to W3, W3 back to W1, and W4 to W1 and W3.

Definition

A link has a **direction**, so draw it as an arrow. The adjacency matrix records it: $a_{ij} = 1$ if page i **links to** page j, 0 if not. Row = from, column = to.

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$



What the matrix shows

- **Not symmetric**: W1 links to W2, but W2 does not link back, so $a_{12} = 1$ while $a_{21} = 0$.
- **Row sums** count links **out**; **column sums** count links **in**. W3 has 3 links in; W4 has none.

Squaring Counts Two-Click Paths

The same product rule, $(A^2)_{ij} = \sum_k a_{ik}\, a_{kj}$, now follows the arrows: a term is 1 exactly when $i \to k$ and $k \to j$.

$$A^2 = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

Check one entry

$$(A^2)_{41} = \underbrace{\begin{bmatrix} 1 & 0 & 1 & 0 \end{bmatrix}}_{\text{row 4 of } A} \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}}_{\text{column 1}} = 1$$

One path: W4 $\to$ W3 $\to$ W1.

What it says

- $(A^2)_{ij}$ **is the number of two-click paths** from page i to page j.
- A^2 **is not symmetric either**: $(A^2)_{41} = 1$ but $(A^2)_{14} = 0$.
- **Column 4 is still all zeros**: no path of any length reaches W4, because nothing links to it.

Higher Powers

The general fact

$(A^n)_{ij}$ counts the **walks of length** n from i to j: sequences of n connections that start at i and end at j. It follows from the same argument applied one step at a time, since $A^n = A \cdot A^{n-1}$.

Two things to keep straight:

- A walk may **revisit** people: $(A^3)_{12}$ includes P1 to P2 to P1 to P2. For routes that never repeat a person, powers overcount.
- The entries **grow fast**. How fast, and which entries dominate, is what **eigenvalues** measure (Unit 2-6), and it underlies Google's PageRank: next slide.

What Else Linear Algebra Does with Networks

With today's tools

- **Triangles**: $\frac{1}{6}\,\mathrm{tr}(A^3)$ counts friend groups of three (tr sums the diagonal). Our Facebook network has 1.
- **Distance**: the smallest n with $(A^n)_{ij} > 0$. P2 and P4 are 2 steps apart.
- **Random surfing**: divide each row of A by its sum; its powers give the chance of being on each page after n clicks.

With eigenvectors (Unit 2-6)

- **Influence**: $A\bar{x} = \lambda\bar{x}$ scores each person by their friends' scores (**eigenvector centrality**).
- **PageRank**: where random surfing settles in the long run, an eigenvector of the link matrix.
- **Communities**: eigenvectors of the **Laplacian** $L = D - A$ (degrees on D's diagonal) split a network into tight-knit groups.

And in AI: a **graph neural network** layer multiplies by A, so each node mixes in its neighbors' features.

Review: Matrices

What we covered: the object and its arithmetic

1. "Is $\bar{b}$ a combination of these vectors?" is $A\bar{x} = \bar{b}$ with the vectors as the **columns** of A. A solution is the list of weights; an impossible equation like $0 = 2$ means $\bar{b}$ is out of reach.
2. An $m \times n$ **matrix** is a grid with m rows and n columns; the entry a_{ij} is in row i, column j. Rows first, always. A column vector is an $n \times 1$ matrix.
3. A **data table** is a matrix: rows are records, columns are features, and both views matter.
4. Know on sight the **zero** and **identity** matrices.
5. **Addition** and **scalar multiplication** are entry by entry and need matching shapes.
6. The **transpose** swaps rows and columns, $(A^\top)_{ij} = a_{ji}$. A is **symmetric** when $A^\top = A$, and $A + A^\top$ always is.

Review: Matrices

What we covered: multiplication and powers

7. Two views of $A\bar{x}$: the **column view**, a linear combination of A's columns weighted by $\bar{x}$, and the **row view**, one **dot product** per entry. A system is now one equation, $A\bar{x} = \bar{b}$.
8. $I\bar{x} = \bar{x}$ and $0\,\bar{x} = \bar{0}$: the identity leaves every vector alone, the zero matrix sends every vector to the origin.
9. $(AB)_{ij}$ is **row i of A dotted with column j of B**; the shapes must chain as $(m \times n)(n \times p) = m \times p$. In block notation, $AB = \begin{bmatrix} A\bar{b}_1 & \cdots & A\bar{b}_p \end{bmatrix}$: one matrix-vector product per column of B.
10. "Product" is a name **by analogy**: products are associative and distributive, $(AB)^\top = B^\top A^\top$, but the shapes must fit and $AB \neq BA$ in general.
11. **Powers** A^n need a square A and are not entry by entry. In a network, $(A^n)_{ij}$ counts the n-**step walks** from i to j.

Notebooks: `U2-3_Matrices-1`, `-2`. **Previous:** Unit 2-2. **Next:** Unit 2-4, **Transformations**.